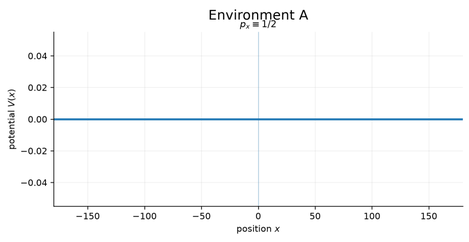
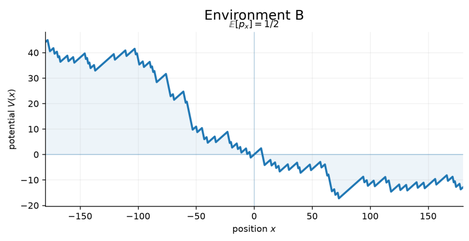
1D Random Walk in Random Environment

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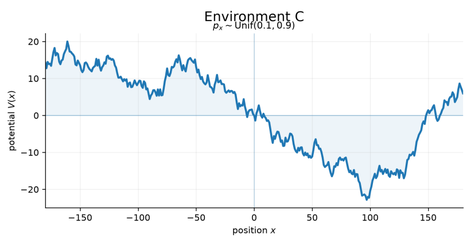
1. A seemingly fair environment



$$p_x \equiv \frac{1}{2}, \rho_x = \frac{1-p_x}{p_x} = 1, \qquad V(n) = \sum_{i=1}^{n} \log \rho_i \equiv 0$$

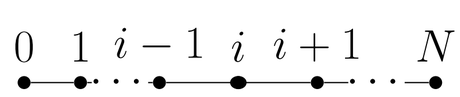


$$p_x = \begin{cases} 0.9, \text{w.p. } 0.2 \\ 0.4, \text{w.p. } 0.8 \end{cases} \qquad E[p_x] = \frac{1}{2}, E[\log \rho_x] < 0$$



$$p_x \overset{\text{i.i.d}}{\sim} \text{Uniform}(0.1, 0.9) \qquad E[p_x] = \frac{1}{2}, E[\log \rho_x] = 0$$

2. Hitting Prob. in a Fixed Env.



$u_i := P_i(T_0 < T_N)$: Reaching 0 before N

$u_i = p_i u_{i+1} + (1-p_i)u_{i-1}, u_N = 0, u_0 = 1$

$p_i(u_{i+1} - u_i) = (1-p_i)(u_i - u_{i-1})$: telescoping

$$u_i = \frac{\sum_{n=i}^{N-1} \prod_{m=1}^{n} \rho_m}{\sum_{n=0}^{N-1} \prod_{m=1}^{n} \rho_m} \qquad \left(\prod_{m=1}^{0} \rho_m := 1 \right)$$

3. The i.i.d Random Environment

$p_i \overset{\text{i.i.d.}}{\sim} \mu$: fixed the environment

$$\prod_{i=1}^{n} \rho_i = \exp\left(\sum_{i=1}^{n} \log \rho_i \right)$$

$$\frac{1}{n} \sum_{i=1}^{n} \log \rho_i \xrightarrow{\text{a.s.}} \mathbb{E}[\log \rho]$$

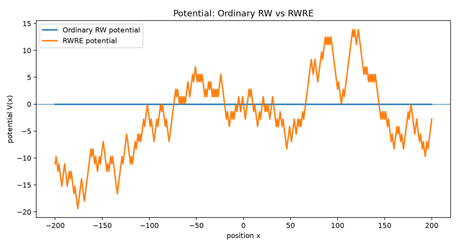
$$\begin{aligned} \mathbb{E}[\log \rho] < 0 &\Longrightarrow \text{right transient}, \\ \mathbb{E}[\log \rho] = 0 &\Longrightarrow \text{recurrent}, \\ \mathbb{E}[\log \rho] > 0 &\Longrightarrow \text{left transient}. \end{aligned}$$

4. Potential and Trapping

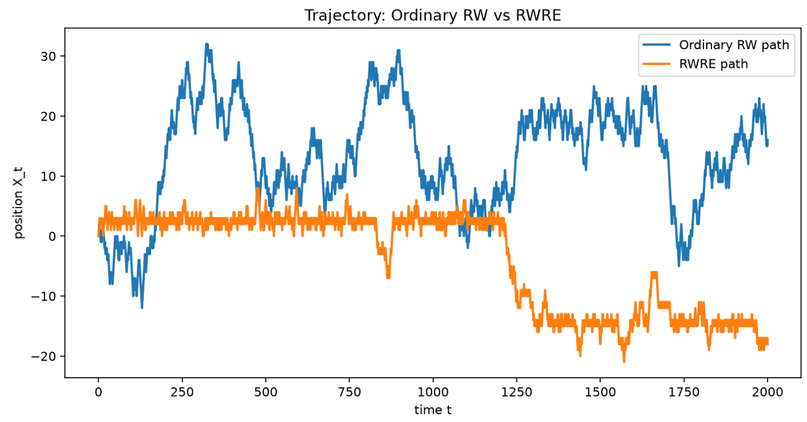
$$V(n) := \sum_{i=1}^{n} \log \rho_i, \qquad V(n) - V(n-1) = \log \rho_n,$$

$$p_n > \frac{1}{2} \iff \log \rho_n < 0, \qquad p_n < \frac{1}{2} \iff \log \rho_n > 0.$$

V acts like a potential: valleys can trap the walk.



$$p_x = \begin{cases} 0.8, \text{w.p. } 0.5 \\ 0.2, \text{w.p. } 0.5 \end{cases}$$



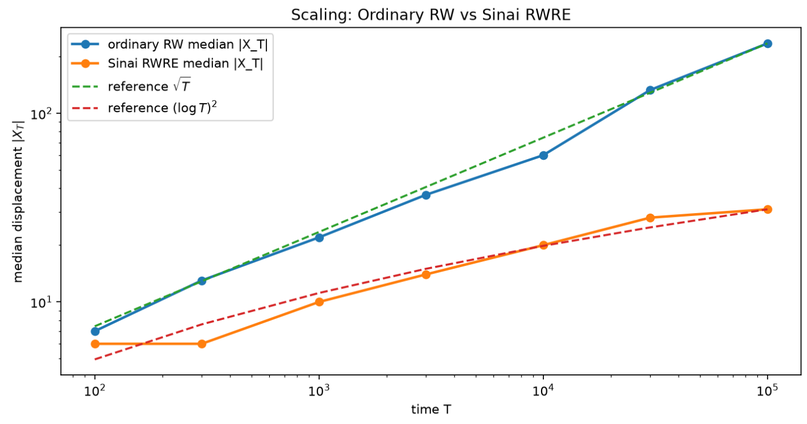
5. Sinai Walk & Scaling

$$\mathbb{E}[\log \rho] = 0, \quad 0 < \text{Var}(\log \rho) < \infty$$

Simple random walk: $|X_t| \asymp \sqrt{t}$

Sinai walk: $|X_t| \asymp (\log t)^2$

$$(\log t)^2 \ll \sqrt{t}$$



6. Takeaways & Resources

- $\mathbb{E}[p_x] = 1/2$ does not imply fairness.
- $\text{sign}\, \mathbb{E}[\log \rho]$ determines the long-term behavior.
- Trapping slows displacement from $\sqrt{t}$ to $(\log t)^2$.

Doyle–Snell (1984);Solomon (1975); Sinai (1982).

Code & references